

UIL POLISH NOTATION EXAMPLES

Operators: +, -, *, /, ^(power)

INFIX

$3 + 4 * 5 - 6$

$(3 + 4) * (5 - 6)$

PREFIX

$- + 3 * 4 5 6$

$* + 3 4 - 5 6$

POSTFIX

$3 4 5 * + 6 -$

$3 4 + 5 6 - *$

Infix to Postfix

Fully Parenthesized the Expression. An FPE has exactly one set of Parentheses enclosing each operator and its operands.

$(A + B) * C - (D + E) / F \rightarrow (((A + B) * C) - ((D + E) / F))$

Move each operator to the right of its operands and remove the parentheses:

$((((A + B) * C) - ((D + E) / F)))$

$A B + C * D E + F / -$

- Operand order does not change!
- Operators are in order of evaluation!

IMPORTANT POINTS

Only the operators move, never the operands.

Notice how in this example the **addition operator** is either before or between or after its **operands**, which are A and the product of B and C. In every notation A is before B and C. The operands don't move, only the operator:

PREFIX

+ [A] [* B C]

INFIX

[A] + [B * C]

POSTFIX

[A] [B C *] +

Don't rewrite infix in normal math notation:

- Leave the operators and operands as they are.
- Leave an expression like $5 + -4$ as is. Don't change it to $5 - 4$.
- Leave $(x + 2)/3$ as is. Don't rewrite as $\frac{x+2}{3}$
- Don't rewrite $2 \wedge 3 \wedge 2$ as 2^{3^2}

Note that exponentiation is right associative, which means:

$$2^{3^2} = 2 \wedge 3 \wedge 2 = 2 \wedge (3 \wedge 2) = 2^{(3^2)} = 2^9 = 512$$

$$\text{Not: } (2 \wedge 3) \wedge 2 = (2^3)^2 = 8^2 = 64$$

Prefix = PN = Polish Notation

Postfix = RPN = Reverse Polish Notation